

Curiosity Rotation Syllabus

Two structured reading blocks + Blender + loose essay practice

Rotation (reading blocks)	Thu 16:30–17:30 & Sun 11:00–12:00
Blender	Sat 18:00–19:30
Drawing / writing	Mon & Tue eves & Sat morning
Essay writing	folded into rotation slots, ~1 per block

Block A – The History of Life on Earth (~8 weeks)

Your dinosaurs / deep-time / historical-animals umbrella: life on Earth as a single narrative, shaped by a constantly changing planet. (Formal names: paleobiology, historical biogeography, deep time.)

Backbone – structured course

- **Dino 101: Dinosaur Paleobiology** (Coursera, University of Alberta) – free, 12 lessons, beginner level. Video lectures from museums & dig sites, with quizzes after each lesson. Its set textbook is Brusatte's *Dinosaur Paleobiology*, so your reading feeds straight into it.
- **Follow-ons** (same university, if you want to continue): Ancient Marine Reptiles; Theropod Dinosaurs & the Origin of Birds; Early Vertebrate Evolution.

Written / reading spine

- **The Rise and Fall of the Dinosaurs** – Steve Brusatte. Narrative, beginner-friendly; pairs with Dino 101.
- **A Short History of Nearly Everything** – Bill Bryson. The deep-time / shifting-continents / "humans are a blip" wonder.
- **The Rise and Reign of the Mammals** – Steve Brusatte. Continues the story past the dinosaurs.

Watch (supplementary)

- **PBS Eons** (YouTube, youtube.com/@eons) – standalone ~10-min episodes; pick ones matching your current topic. A fan playlist orders them by deep time: search "PBS Eons Deep Time Playlist".
- **Prehistoric Planet** (Apple TV+) – for the visceral "what was it actually like" hit.

Suggested arc (video → written → applied)

- **Weeks 1–2:** Dino 101 lessons 1–3 + matching Brusatte chapters; do the lesson quizzes as your "applied" step.
 - **Weeks 3–4:** Dino 101 lessons 4–7; Eons episodes on extinctions.
 - **Weeks 5–6:** Dino 101 lessons 8–12; Bryson deep-time chapters (geology, plate tectonics).
 - **Weeks 7–8:** Begin Rise & Reign of the Mammals; pick an essay thread (see prompts).
- ★ Block goal: a confident mental timeline of life on Earth, and a felt sense of how a changing planet drove what lived, thrived, and vanished.

Block B – Conceptual Physics & Space (~8 weeks)

Understanding-first, maths-light. Genuinely grasp the big ideas (motion, energy, entropy, relativity, cosmology) conceptually. Maths is optional, added later only if

a specific concept makes you want it.

Backbone – structured course

- **Khan Academy – Physics** (khanacademy.org) – free, structured units, each combining short videos + written articles + interactive practice. That's your video/written/applied loop in one place.
- Start with **Middle School Physics** (conceptual, maths-light: Motion & Forces, Energy, Waves), then step up to **Physics 1** when comfortable. For space: the **Cosmology & Astronomy** unit.

Written / reading spine

- **Conceptual Physics** – Paul Hewitt. The maths-light textbook standard; your written reference.
- **Cosmos** – Carl Sagan. Unmatched awe-plus-understanding; pair with the original series.
- **Astrophysics for People in a Hurry** – Neil deGrasse Tyson. Short, accessible space primer.

Watch (supplementary)

- **PBS Space Time – "Curved Spacetime in General Relativity"** playlist (12 videos) for the relativity weeks. Early episodes build intuition with no maths; later ones get rigorous – slow down rather than push.
- **Veritasium & Sixty Symbols** (YouTube) – single-concept deep dives for the "go deeper" moments.

Suggested arc (video → written → applied)

- **Weeks 1–2:** Khan MS Physics – Motion & Forces (videos + articles + practice) and matching Hewitt chapters.
- **Weeks 3–4:** Khan – Energy; entropy concepts; Veritasium for intuition.
- **Weeks 5–6:** Relativity conceptually – PBS Space Time relativity playlist (no maths required).
- **Weeks 7–8:** Khan Cosmology & Astronomy unit; Sagan / Tyson; pick an essay thread.

Note: the Feynman Lectures are wonderful but a maths-heavy undergraduate course in disguise – a destination, not a start. Save as an "if I get hooked" reward.

★ Block goal: comfortably follow and discuss big physics ideas conceptually – and know which one you'd want to unlock with maths, if any.

Blender – Weekly Making-Pursuit (ongoing)

Sat 18:00–19:30, every week. A continuous project track, NOT a rotation subject – last week's progress carries into this week. Builds 3D + creative-confidence skills now, and feeds game dev later if that pull strengthens.

Backbone – structured course

- **CG Fast Track – Blender Track** (you own this). Continue along the beginner→intermediate→advanced path you're already on.
- **You are here:** Rendering in Cycles (Intermediate). Next up the track: Compositing → Hard Surface Boot Camp Vols 1–3 → Procedural Shaders → Organic Modeling → Sculpting → Rigging → Geometry Nodes.

Written companion (the "why" behind the videos)

- **Blender official docs** (docs.blender.org) – free, always current, organised by the same workflow categories as CG Fast Track. Read the matching section *after* each module (e.g. after Rendering in Cycles, read the Cycles docs).
- **The Complete Guide to Blender Graphics** – John M. Blain. A book-shaped comprehensive reference if you want one later; updated per Blender version.

Workflow per session: watch the CG Fast Track segment → read the matching docs section → apply it in your own file. The key to 1.5 hrs/week working is keeping it weekly – continuity beats hours; skipping weeks is what kills slow Blender learning.

Essay Writing – Loose Practice

Target: ~one short essay (500–1000 words) per rotation block, tied to what you've been reading. The reading is the fuel – you're already making connections, the essay is where you write them down. Any prompt is abandonable the instant your own idea takes over; the prompt only exists to beat the blank page.

Starter prompts – History of Life

- What does "deep time" do to your sense of human significance? Humbling, freeing, or unsettling?
- Pick one extinction event. What's most surprising about what survived vs what didn't?
- If continents hadn't drifted, how different might life look? Use one concrete example.

Starter prompts – Physics & Space

- Which physics concept most changed how you see everyday life? Trace exactly why.
- Is the maths-free understanding you've built "real" understanding? Argue it both ways.
- Sagan's "pale blue dot" – does the cosmic perspective console or diminish? Make your case.

Keep it lit-essay in spirit: a claim, evidence, and the connections you missed making – not a book report.

Block A tracker – History of Life

Material	Type	Done
Dino 101 – lessons 1–12 (Coursera)	Course	☐
Rise and Fall of the Dinosaurs – Brusatte	Read	☐
A Short History of Nearly Everything – Bryson	Read	☐
Rise and Reign of the Mammals – Brusatte	Read	☐
PBS Eons – episodes matching topics	Watch	☐
Essay: one short piece (500–1000 words)	Write	☐

Block B tracker – Conceptual Physics & Space

Material	Type	Done
Khan Academy – MS Physics + Physics 1 units	Course	☐
Khan Academy – Cosmology & Astronomy unit	Course	☐
Conceptual Physics – Hewitt	Read	☐

Cosmos – Sagan	Read	☐
PBS Space Time – relativity playlist (12)	Watch	☐
Essay: one short piece (500–1000 words)	Write	☐

Blender tracker – CG Fast Track

Material	Type	Done
Rendering in Cycles (in progress)	Course	☐
Compositing Fundamentals	Course	☐
Hard Surface Boot Camp Vols 1–3	Course	☐
Procedural Shader Networks	Course	☐
Organic Modeling / Sculpting	Course	☐
Blender docs – read matching section per module	Read	☐